MATHEMATICAL INDUCTION

Axioms of Peano Arithmetic

- 1 is a natural number
- If x is a natural number, then x + 1 is also a natural number
- There is no natural number, Z, such that Z + 1 = 0.
- Given natural numbers x and y, if x + 1 = y + 1, then x = y.
- If one can prove that if a property holds for some natural number x then it holds for x + 1, and if it can further be proven that the same property holds for 1, then the property holds for all natural numbers.

Principle of Mathematical Induction

- Suppose that for each positive integer n we have a statement S(n) that is either true or false. Suppose that

  S(1) is true; and (1)

  if S(i) is true, for all $i \leq n$, then S(n + 1) is true. (2)

- Then S(n) is true for every positive integer n.
- Condition (1) is sometimes called the **Basis Step** and condition (2) is sometimes called the **Inductive Step**.

Some Remarks on the Principle of Mathematical Induction

- The axiom allows us to prove that a certain property holds for all natural numbers.
- The idea is that, in order to prove that a formula (or theorem) about natural numbers is true for all natural numbers, we first verify (by actual substitution) that the formula is true for the number 1.

Some Remarks on the Principle of Mathematical Induction

- Then, we prove that whenever the formula is true for a natural number n, it must also be true for $n + 1$.
- Since it was verified to be true for 1, then it had to be true for 2; and since it was true for 2, it had to be true for 3, and so on for all natural numbers.

Principle of Mathematical Induction: Example 1

- Prove by mathematical induction that

$$\sum_{i=1}^{n} i = \frac{n(n+1)}{2}$$

- Solution.

  Let S_n denote the sum of the first n positive integers

  S_n = =1 + 2 + 3 + . . . + n.

  Here, a sequence of statements is actually being made, namely,

$$S_1 = 1 = \frac{1(2)}{2} \qquad S_2 = 1 + 2 = \frac{2(3)}{2} = 3$$

$$S_3 = 1 + 2 + 3 = \frac{3(4)}{2} = 6$$

Principle of Mathematical Induction: Example 1

- The formula is true for the first 3 natural numbers. Assuming that all of the equations preceding the (n + 1)st equation are true, then, in particular, the nth equation is true, that is,

$$S_n = 1 + 2 + 3 + ... + n = \frac{n(n+1)}{2}$$

- We must show that the (n + 1)st equation

$$S_{n+1} = \frac{(n+1)(n+2)}{2}$$

is true. Take note that

$$S_{n+1} = 1 + 2, + . . . + n + (n + 1).$$

Principle of Mathematical Induction: Example 2

- Use mathematical induction to show that

  $n! \geq 2^{n-1}$ for $n = 1, 2, \ldots.$

- Proof:

  Basis Step. We must show that the inequality is true if $n = 1$. This is easily accomplished, since

  $1! = 1 \geq 1 = 2^{1-1}.$

  Inductive Step. We must show that if $i! \geq 2^{i-1}$ for $i = 1, \ldots, n$, then

  $(n + 1)! \geq 2^{n}.$

Principle of Mathematical Induction: Example 2

- To show that if $i\,! \geq 2^{i-1}$ for $i = 1, \ldots, n$, then $(n + 1)! \geq 2^n$

we assume that $i! \geq 2^{i-1}$ for $i = 1, \ldots, n$. Then, in particular, for $i = n$, we have $n! \geq 2^{n-1}$.

We can relate the 2 inequalities above by observing that

$$(n + 1)! = (n + 1)(n!).$$

Now

$$\begin{aligned}(n + 1)! &= (n + 1)(n!) \\ &\geq (n + 1)2^{n-1} \\ &\geq 2 \,.\, 2^{n-1} && \text{since } n + 1 \geq 2 \\ &= 2^n.\end{aligned}$$

Principle of Mathematical Induction: Example 2

- Since we have shown that if $i\,! \geq 2^{i-1}$ for

  $i = 1, \ldots, n$, then $(n + 1)! \geq 2^n$

  therefore, $n! \geq 2n\text{-}1$ for $n = 1, 2, \ldots$ is true.
- We have completed the Inductive Step.
- Since the Basis Step and the Inductive Step have been verified, the Principle of Mathematical Induction tells us that $n! \geq 2^{n-1}$ is true for every positive integer n.

Strong Form of Mathematical Induction

- The verification of the Inductive Step where we assume that $S(i)$ is true for all $i < n + 1$ and then prove that $S(n + 1)$ is true.
- This formulation of mathematical induction is called strong form of mathematical induction.
- Often, as was the case in the preceding examples, we can deduce $S(n + 1)$ assuming only $S(n)$.
- indeed, the Inductive Step is often stated:

If $S(n)$ is true, then $S(n + 1)$ is true.

Principle of Mathematical Induction: Example 3

- Use induction to show that if r $\neq$ 1,

$$a + ar^1 + ar^2 + \ldots + ar^n = \frac{a(r^{n+1} - 1)}{r - 1}$$

  for n = 0, 1, ...
- The sum on the left is called a **geometric sum**. In a geometric sum, the ratio of consecutive terms ($ar^{i+1}/ar^{i} = r$) is constant.
- Proof:

  Basis Step. The Basis Step, which in this case is obtained by setting n = 0, is

  which is true.

Principle of Mathematical Induction: Example 3

- Inductive Step . Assume that statement is true for n. Now,

$$a + ar^1 + ar^2 + \ldots + ar^n + ar^{n+1} = \frac{a(r^{n+1} - 1)}{r - 1} + ar^{n+1}$$

$$= \frac{a(r^{n+1} - 1)}{r - 1} + \frac{ar^{n+1}(r - 1)}{r - 1}$$

$$= \frac{a(r^{n+2} - 1)}{r - 1}$$

Principle of Mathematical Induction: Example 3

- Since the modified Basis Step and the Inductive step have been verified, the Principle of Mathematical Induction tells us that

$$a + ar^1 + ar^2 + \ldots + ar^n = \frac{a(r^{n+1} - 1)}{r - 1}$$

is true for n = 0, 1, ...

Principle of Mathematical Induction: Example 3

- As an example of the use of the geometric sum, if we take *a* = 1 and *r* = 2 in the above expression, we obtain the formula

$$1+2+2^2+2^3+\ldots+2^n=\frac{2^{n+1}-1}{2-1}=2^{n-1}-1.$$

Principle of Mathematical Induction: Example 3

The reader has surely noticed that in order to prove the previous formulas one has to be given the correct formulas in advance. A reasonable question is: How does one come up with the formulas? There are many answers to this question. One technique to derive a formula is to experiment with small values and try to discover a pattern. For example, consider the sum $1 + 3 + \ldots + (2n - 1)$. The following table gives the values of this sum for $n = 1, 2, 3, 4$.

Principle of Mathematical Induction: Example 3

n	**1 + 3 + . . . + (2n –1)**
1	1
2	4
3	9
4	16

Since the second column consists of squares, we *conjecture* that

$$1 + 3 + . . . + (2n - 1) = n^2$$

for every positive integer *n*.

The conjecture is correct and the formula can be proved by mathematical induction (see Exercise Set 3.4 #1).

Principle of Mathematical Induction: Example 4

- Use induction to show that $5^n - 1$ is divisible by 4 for $n = 1, 2, \ldots$.
- **Proof:**

  Basis Step.

  If $n = 1$, $5^n - 1 = 5^1 - 1 = 4$, which is divisible by 4.

  Note that also $4 = 5^0 - 1 + 4 \cdot 5^0$

  If $n = 2$, $5^2 - 1 = 25 - 1 = 24$, which is divisible by 4.

  Also $24 = 5^1 - 1 + 4 \cdot 5^1$

Principle of Mathematical Induction: Example 4

If $n = 3$, $5^{3} - 1 = 125 - 1 = 124$, which is divisible by 4.

Again $124 = 5^2 - 1 + 4 \cdot 5^2$

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If we look at the pattern, we can say that

$$5^{n} - 1 = 5^{n-1} - 1 + 4 \cdot 5^{n-1}.$$

Principle of Mathematical Induction: Example 4

Inductive Step.

Assume that $5^n - 1$ is divisible by 4 and that the above formula holds. We must show that $5^{n+1} - 1$ is divisible by 4. To relate the (n + 1)st case to the nth case, we write

$$5^{n+1} - 1 = (5^n - 1) + 4 . 5^n.$$

Principle of Mathematical Induction: Example 4

By assumption, $5^{n} - 1$ is divisible by 4 and, since $4 \cdot 5^{n}$ is divisible by 4, the sum

$$(5^{n} - 1) + 4 \cdot 5^{n} = 5^{n+1} - 1$$

is divisible by 4. Since the Basis Step and the Inductive Step have been verified, the Principle of Mathematical Induction tells us that $5^n - 1$ is divisible by 4 for $n = 1, 2, \ldots$

Principle of Mathematical Induction: Example 5

- For any set X with n elements, the cardinality of the power set of X has 2^n elements, that is, $n(\wp(X)) = 2^n$. Or

  if $|X| = n$, then $|\wp(X)| = 2^n$

- **Proof**. The proof is by induction on n.

  Basis Step. If $n = 0$, X is the empty set. The only subset of the empty set is the empty set itself; thus

  $$|\wp(X)| = 1 = 2^0 = 2^n.$$

  Thus the formula is true for $n = 0$.

Principle of Mathematical Induction: Example 5

Inductive Step. Assume that formula holds for n. Let X be a set with $|X| = n + 1$. Choose $x \in X$. We divide $\wp(X)$ into two classes. The first class consists of those subsets of X that include x, and the second class consists of those subsets of X that do not include x. If we list the subsets of X that do not include x,

$$X1, X2, \ldots, Xk,$$

then

Principle of Mathematical Induction: Example 5

$$X1 \cup \{x\}, X2 \cup \{x\}, \ldots, Xk \cup \{x\}$$

is a list of the subsets of X that do include x. Thus the number of subsets of X that include x is equal to the number of subsets of X that do not include x. If $Y = X - \{x\}$, $\wp(Y)$ consists precisely of those subsets of X that do not include x. Thus $|\wp(X)| = 2|\wp(Y)|$. Since $|Y| = n$, by the inductive assumption, $|\wp(Y)| = 2n$.

Principle of Mathematical Induction: Example 5

- Therefore,

$$|\wp(X)| = 2|\wp(Y)| = 2 \cdot 2n = 2n+1.$$

- Thus the formula holds for $n + 1$ and the inductive step is complete. By the Principle of Mathematical Induction, Law 3.6.2 holds for all $n \geq 0$.